

WINDOW ROCK, AZ, USA

WMO: 722764

Lat: 35.658N

Lon: 109.061W

Elev: 6739

StdP: 11.45

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 03029

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.6	(c) 6.1	(d) -9.2	(e) 4.3	(f) 20.4	(g) -5.8	(h) 5.2	(i) 24.8	(j) 25.0	(k) 42.5	(l) 21.8	(m) 39.7	(n) 0.9	(o) 190	(p) 0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 30.5	(c) 90.5	(d) 56.4	(e) 88.2	(f) 55.9	(g) 86.0	(h) 55.3	(i) 61.9	(j) 76.3	(k) 60.8	(l) 75.5	(m) 59.9	(n) 74.7	(o) 9.8	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.0	92.8	64.0	56.8	88.6	63.9	55.2	83.4	64.1	30.9	76.0	30.1	75.2	29.4	74.0	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.4	(b) 20.9	(c) 18.2	(e) -8.7	(f) 94.3	(g) 6.3	(h) 2.1	(i) -13.2	(j) 95.8	(k) -16.9	(l) 97.0	(m) -20.4	(n) 98.2	(o) -25.0	(p) 99.6		
(a) 25.4	(b) 20.9	(c) 18.2	(e) -9.3	(f) 64.3	(g) 6.1	(h) 1.6	(i) -13.6	(j) 65.4	(k) -17.2	(l) 66.3	(m) -20.6	(n) 67.2	(o) -25.1	(p) 68.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	29.3	33.3	40.3	46.7	54.7	65.4	70.4	67.5	60.4	49.1	38.4	29.6		
	DBStd	15.58	8.08	7.60	6.44	6.13	6.54	5.13	3.26	3.27	5.13	6.35	7.09	7.85		
	HDD50	2653	642	467	305	129	31	0	0	0	3	92	351	633		
	HDD65	6242	1107	887	767	548	321	56	2	14	150	493	799	1098		
	CDD50	2230	0	0	3	32	178	461	633	542	316	63	2	0		
	CDD65	343	0	0	0	0	3	67	170	91	12	0	0	0		
	CDH74	7329	0	0	0	15	378	2244	2461	1544	645	42	0	0		
	CDH80	2564	0	0	0	0	66	883	1006	480	127	2	0	0		
(14) Wind	WSAvg	5.6	4.2	5.6	6.5	8.2	7.6	6.9	5.1	4.4	4.8	5.0	4.7	4.4		
(15) Precipitation	PrecAvg	9.9	0.7	0.8	0.7	0.5	0.5	0.2	1.3	1.7	1.1	0.9	0.6	0.8		
	PrecMax	14.4	3.4	3.7	2.2	1.4	2.9	0.9	3.8	3.6	3.3	2.9	1.7	2.2		
	PrecMin	4.8	0.1	0.1	0.0	0.1	0.0	0.0	0.1	0.1	0.3	0.0	0.0	0.1		
	PrecStd	2.3	0.7	0.7	0.6	0.4	0.6	0.2	0.9	0.9	0.8	0.8	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.4	64.1	71.9	76.3	85.9	93.8	93.3	90.5	86.8	78.8	69.8	58.9		
		MCWB	40.3	41.9	45.8	47.4	53.2	55.5	57.6	57.0	55.6	51.0	45.7	40.1		
	2%	DB	54.1	60.2	67.4	72.3	81.4	90.5	90.9	87.7	83.6	74.5	66.0	54.6		
		MCWB	38.2	40.4	43.8	45.7	51.1	54.7	57.4	56.9	54.3	49.2	43.7	38.3		
	5%	DB	50.0	54.6	63.7	69.8	78.6	88.0	88.5	85.1	81.0	71.7	61.8	50.3		
		MCWB	36.1	37.7	42.1	44.7	49.1	53.8	57.1	56.3	54.0	48.3	42.3	36.8		
	10%	DB	45.4	49.7	59.7	66.3	75.0	85.4	86.1	82.4	78.1	68.2	57.1	45.6		
		MCWB	34.5	35.9	40.1	43.2	47.9	52.6	57.2	56.9	53.4	47.0	40.1	34.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	44.4	46.9	49.0	55.1	59.5	63.4	63.2	60.2	55.7	47.3	43.3		
		MCDB	54.2	58.2	68.1	71.2	81.9	79.8	77.9	75.4	72.6	65.7	62.4	53.0		
	2%	WB	39.6	41.3	44.5	47.0	52.4	57.9	62.2	61.8	58.5	52.5	45.4	40.4		
		MCDB	50.6	55.4	64.5	69.3	76.4	80.0	77.4	75.2	72.7	67.0	61.0	50.4		
	5%	WB	37.3	39.2	42.8	45.4	50.6	56.5	61.1	60.8	57.3	50.6	43.4	37.9		
		MCDB	46.0	52.0	61.0	67.0	73.6	80.1	76.7	74.5	72.2	65.4	59.3	47.8		
	10%	WB	35.3	37.2	41.1	43.9	49.1	54.9	60.2	59.9	56.1	48.7	41.4	35.5		
		MCDB	44.1	47.9	57.5	64.4	71.5	79.8	75.7	73.6	71.3	64.3	56.2	45.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	29.4	28.0	31.8	32.9	34.1	37.3	30.5	29.7	32.7	33.4	32.9	29.0		
		MCDBR	36.2	37.6	41.1	41.0	40.4	41.4	35.7	35.8	38.2	40.4	41.7	36.3		
		MCWBR	22.3	21.2	21.2	20.3	18.1	17.2	12.6	12.3	15.5	20.0	23.3	22.5		
	5% WB	MCDBR	30.5	31.9	36.6	37.0	36.6	35.9	27.1	26.1	28.5	31.8	35.6	31.0		
		MCWBR	19.4	18.5	19.1	18.3	16.6	15.0	8.8	8.6	11.8	16.3	20.3	19.7		
(40) Clear-Sky Solar Irradiance	taub		0.234	0.242	0.261	0.277	0.288	0.299	0.343	0.336	0.295	0.278	0.243	0.237		
	taud		2.645	2.627	2.567	2.506	2.518	2.476	2.439	2.423	2.547	2.555	2.623	2.634		
	Ebn at Noon		318	325	324	320	315	310	296	297	307	305	310	310		
	Edh at Noon		22	25	30	33	33	35	36	36	30	27	22	21		
(44) All-Sky Solar Radiation	RadAvg		988	1287	1761	2180	2454	2627	2238	2050	1838	1473	1115	870		
	RadStd		89	110	111	111	166	118	120	78	65	113	68	61		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page